Infectious diseases – AS 9700 Biology Nov 2022

1. Nov/2022/Paper_11/No.38

Penicillin is an antibiotic used to treat infections caused by bacteria.

Which events explain how bacteria become resistant to penicillin?

1 The number of bacteria with the allele for resistance to penicillin increases. – plasmids help exchange genes of resistance

2 Resistance to penicillin is the result of a mutation in a bacterium. – change in alleles

3 Bacteria with the allele for resistance to penicillin pass the allele to their daughter cells. – through Plasmids

4 The mutation for resistance to penicillin is always caused by the presence of penicillin. Mutations are caused by need to survive the effects of penicilin

A 1, 2, 3 and 4

B 1, 2 and 3 only

C 1 and 4 only

D 2 and 3 only

2. Nov/2022/Paper_11/No.39

Which statements explain why some vaccines can be taken by mouth but tuberculosis (TB) vaccine has to be injected?

1 Macrophages present antigens in vaccines to stimulate an immune response. macrophages are not in vaccines

2 The TB antigens necessary to produce an immune response are proteins which would be digested in the stomach and small intestine. vaccines have weakened antigens – antigens are made of proteins Proteins are digested at stomach

3 There are no B-lymphocytes and T-lymphocytes in the stomach.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 only

3. Nov/2022/Paper_12/No.38

Cholera and tuberculosis are infectious diseases that can spread when people have to live in overcrowded conditions.

Which strategies would help control the spread of cholera **and** tuberculosis?

1 covering food to prevent flies landing on it – for cholera control only

2 providing chlorinated drinking water – for cholera control only.

3 treating severe cases with antibiotics – both TB and cholera are caused by bacteria

A 1, 2 and 3 **B** 1 only **C** 2 and 3 only **D** 3 only

4. Nov/2022/Paper_21/No.4(b)

(b) The glycoproteins haemagglutinin and neuraminidase are found in the influenza virus and in the virosomes used in a vaccine against the influenza virus.

Haemagglutinin binds to a receptor in the cell surface membrane of phagocytes.

Suggest why haemagglutinin is present in virosomes used in the vaccine for influenza.

It acts as a foreign antigen. It stimulates the primary immune response. This provides artificial immunity leading to formation of antigen presenting cell. Haemaagglutinin stimulates clonal expansion of lymphocytes. The lymphocytes destroy antigens by phagocytosis. Then memmory cells remain in circulation [2]

5. Nov/2022/Paper_21/No.5d(ii)

(ii) Malaria is another infectious disease that also involves an insect vector.

State the type of organism that causes malaria **and** name **one** organism that causes malaria in humans.

type of organism ..... protoctist .....

name ..... Plasmodium falciparum .....

[2]

6. Nov/2022/Paper_23/No.2(b_ c)

(b) Fig. 2.2 shows the reported number of new cases of tuberculosis (TB) in the USA and the number of new cases per 100 000 of the population of the USA between 1993 and 2018.

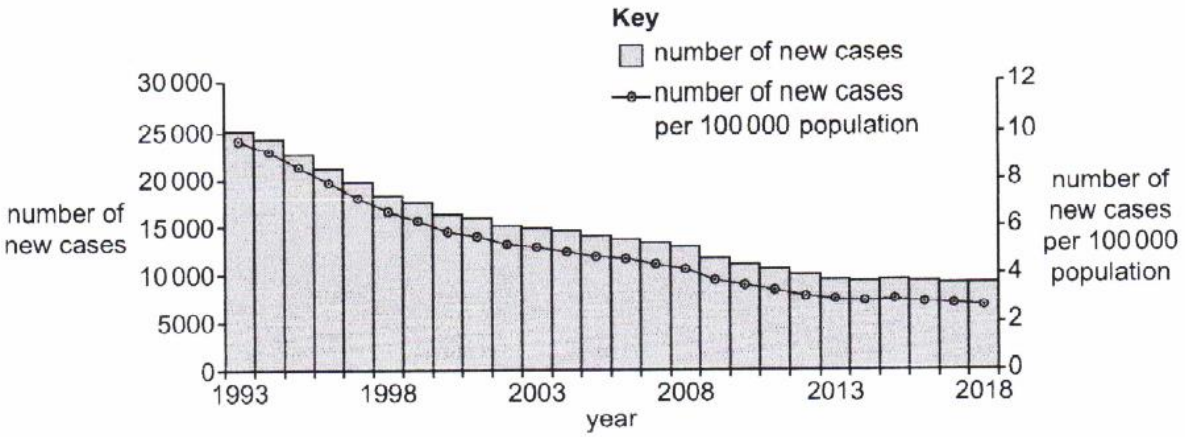


Fig. 2.2

(i) Calculate the percentage change in the number of new cases of TB in the USA between 1993 and 2018.

Show all your working.

New cases in 1993 = 25000

New cases in 2018 = 8000

8000 − 25000 = −17000

$\frac{17000}{25000} \times 100 = 68\%$

answer 68 % [3]

(ii) Use Fig. 2.2 to describe the trend in the number of new cases of TB in the USA between 1993 and 2018.

There is a decrease in the number of new cases. There is a sharp decrease between 1993 and 2001. However, the number of new cases is constant between 2013 to 2018. There is an increase in numbers of new cases in 2015. [2]

(c) Suggest an advantage of calculating the number of new cases per 100 000 each year in the prevention and control of TB across the world.

It is more accurate. It allows for comparison between the years and countries.

[1]